In fixed capacitor, capacitance value cannot be changed.

Its symbol is shown in Figure 1.54 (a).

1.11.7.1 Types of fixed Capacitors

1. Paper capacitor
2. Mica capacitor
3. Polyester capacitor
4. Ceramic capacitor
5. Electrolytic capacitor (polarized and non-polarized)
6. Bipolar Electrolytic Capacitor
7. Tantalum capacitors (metal film capacitors)
8. SMD capacitor
9. Ultra capacitor (or) super capacitor

1.11.7.1.1 Paper capacitor

Figure 1.54(b) shows a paper capacitor. In this, wax paper is rolled in the form of cylinder and dipped in wax solution in order to exhaust the air. Then, it is placed in between two thin aluminium plates.

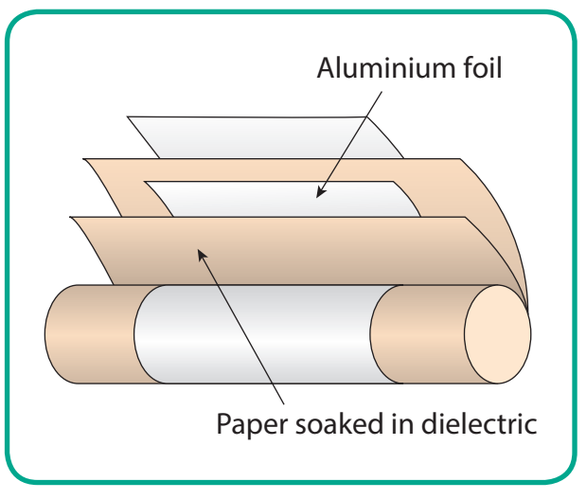


Figure 1.54 (b) Paper Capacitor

Available capacitor values are from 0.001 μF. Voltage rating from 100 V to 1500 V.

Uses

Paper capacitor is used for the following purposes.

1. Blocking
2. Coupling
3. Decoupling
4. Filtering
5. Bypassing

1.11.7.1.2 Mica Capacitor

Figure 1.55 shows a mica capacitor. In this type, mica is used as the dielectric medium. Silver coated on the mica acts as electrodes.

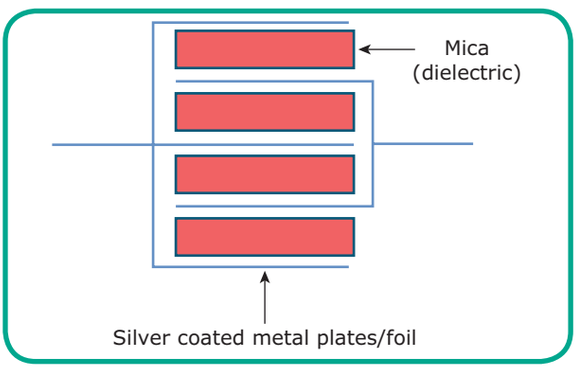


Figure 1.55 Mica Capacitor

This type of capacitor's values are available from 2 pF to 1000 pF and voltage rating from 6V to 500V.

Uses

1. Used in high frequency coupling circuits.
2. Used in tuning circuits.

1.11.7.1.3 Polyester capacitors

Figure 1.56 shows a Polyester capacitor.

It uses polyester film as a dielectric medium and aluminium foils as conducting plates.

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